A120600 concise calculus certificate

VERIFIED

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Definition and first terms

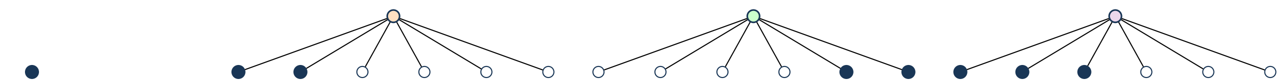
$7\,A(x) = 6 + 1\,x + A(x)^6$, with the origin-normalized solution.

Normalization/grammar: `A(x)=1+(1)*T(x); T=x+15*T^2+20*T^3+15*T^4+6*T^5+1*T^6.`

$a(0), \ldots, a(7) = 1, 1, 15, 470, 18390, 805806, 37828981, 1860433080.$

Kernel used throughout: $\rho(u) = u\,(1 - 15\,u^1 - 20\,u^2 - 15\,u^3 - 6\,u^4 - 1\,u^5)$, so $x = \rho(T(x))$.

Typogeometry



`1x1` `{1,1,0,0,0,0}x1` `{0,0,0,0,1,1}x1` `{1,1,1,0,0,0}x1`

literal 6-slot geometry; empty positions are shown. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2: 15, Δ_3: 20, Δ_4: 15, Δ_5: 6, Δ_6: 1.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \ldots, k_5) \in \mathbb{Z}^5_{\geq 0} : \sum_{j=1}^{5} jk_j = n - 1\}, \quad K = \sum_{j=1}^{5} k_j,$$

$$a(n) = \frac{1}{n} \sum_{\boldsymbol{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \ldots, k_5} 15^{k_1} 20^{k_2} 15^{k_3} 6^{k_4} 1^{k_5}.$$

$$a(n) = \frac{1}{2\pi i n} \oint_\gamma \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^{5} P_r(n) a(n + r) = 0$ for $n \geq 1$. The exact degree-5 coefficient polynomials are embedded in `certificate_payload.json` (source: `case.json#/objects/p_recurrence`).

Differential equation

The scalar linear ODE has order 5. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `case.json#/objects/ode_from_recurrence`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n, u) = N(n, u)/\rho(u)^4$, $\deg_u N = 25$.

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.

Matrix dimensions: 12×12; remainder 5×6, rank 5, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Canonical records: `data/tree_model.json`, `data/contour.json`, `data/matrices.json`, `data/recurrence.json`, `data/ode.json`. Embedded payload contains exact sources and SHA-256 matrix identifiers. Compare with the verbose certificate for A120590.